

Final Project, Part 1 — Speed: The Cost of Execution Delay

Group 06

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Assets: AAPL, GPRO, EUR/USD | **Period:** September 2019, 20 trading days
Signals: moving average and on-balance volume (paper), persistent quote imbalance (ours)
Delays: 0 ms, 100 ms, 1 s | **Book:** \$1,000,000

Summary

We apply the method of Scholtus and van Dijk (2012) to a liquid stock, a less liquid stock and a currency pair. A trading signal is computed every minute and the order it implies is executed 0 ms, 100 ms or 1 second later at the best bid or ask of that moment. The question is how much return the delay costs.

1. **All 27 strategies lose money once the spread is paid**, from -0.19% (EUR/USD, moving average) to -109% (GPRO, imbalance). On GPRO the three signals earn money at the mid quote, and the 22.5 bps spread takes it all back.
2. **For one strategy over one month the cost of delay is too small to measure**. Averaged over the nine asset/signal pairs it is $+0.02\%$ at 100 ms and -0.12% at 1 second, with a 90% interval of -0.42% to $+0.14\%$.
3. **Pooling the orders of a 601-rule universe makes it measurable, at 1 second and not at 100 ms**. Executing 1 second late costs 0.07 bps per order for moving-average rules on AAPL, 0.06 bps for moving-average rules on GPRO and 0.11 bps for imbalance rules on GPRO. At 100 ms every effect is within 0.02 bps.
4. **The paper's central result holds on the liquid stock, after its selection-bias test**. On AAPL, delay lowers the return of profitable rules by 0.7% at 200 ms and 3.1% at 1 second, more than random rules lose ($p = 0.03$ and $p < 0.001$). On GPRO and EUR/USD the losses at 1 second (-1.0% and -1.2%) are what selection alone produces.
5. **Tick frequency sets how often a delayed order meets a different quote** (62% of the time after 1 second on AAPL, 55% on EUR/USD, 1% on GPRO); **tick size relative to price sets how much each miss costs** (22 bps on GPRO). Sensitivity to delay is the product of the two.
6. **A \$1,000,000 order fills at the best quote only on EUR/USD**. It is half the displayed size there, 22 times the displayed size on AAPL and 53 times on GPRO.

What we implemented

The assignment asks for two equities and one FX pair over one month, two signals from the paper plus one of our own, three execution delays, the total return on a \$1,000,000 book for each combination, and the cost of delay. The trading procedure follows section 3.1 of the paper and the

instructions of Discussion Session 06.

- **Signal.** Every 60 seconds a rule computes a position in $\{-1, 0, +1\}$ from the information available at the interval change.
- **Execution.** The order is executed a delay d later against the best quote prevailing at that moment: buys pay the ask, sells receive the bid. The bid-ask spread is the only transaction cost.
- **Trading window.** Signals are acted upon from 09:40 to 15:50 New York time. The book is closed every day at the 15:50:00 price.
- **Size.** Every trade is for the number of units worth \$1,000,000 on the asset's first day: 4,850 AAPL shares, 263,505 GPRO shares, EUR 913,013.
- **Return.** The total return of a strategy is the sum of the simple returns of its round trips, not compounded. For a long position entered at t_1 and closed at t_2 , and for a short one,

$$r^{\text{long}} = \frac{p_{t_2+d}^{\text{bid}} - p_{t_1+d}^{\text{ask}}}{p_{t_1+d}^{\text{ask}}}, \quad r^{\text{short}} = -\frac{p_{t_2+d}^{\text{ask}} - p_{t_1+d}^{\text{bid}}}{p_{t_1+d}^{\text{bid}}}.$$

The dollar P&L is the same sum valued on the fixed number of units.

- **Cost of delay.** Total return with delay d minus total return with $d = 0$, for the same signal. *Negative means the delay hurts.*

Three assets, three signals and three delays give 27 profit-and-loss reports: the 18 the assignment counts (three assets, the two signals taken from the paper, three delays) and 9 for our own signal.

Data and the three markets

Equities. Databento Nasdaq TotalView-ITCH, MBP-1 schema: every trade and every change of the best bid and offer, with nanosecond time stamps, from 09:30 to 16:00 New York time.

EUR/USD. Dukascopy tick data: every change of the best bid and ask with the quoted sizes, with millisecond time stamps. The discussion session points to a currency file on the MFE-4 server; we use a public source so that the FX leg covers the same month, the same 20 days and the same clock window (13:30 to 20:00 UTC) as the equities. The three assets then face the same number of signals.

Regular-session average	AAPL	GPRO	EUR/USD
Mid price	217.90	4.51	1.1011
Quoted spread (bps)	0.73	22.52	0.27
One tick, in bps of the price	0.46	22.18	0.09
One-minute volatility (bps)	5.3	17.0	1.1
Quote price changes per second	8.80	0.04	1.21
Quote changed after 100 ms (%)	25.6	0.2	15.0
Quote changed after 1 s (%)	62.2	1.1	55.0
Units in a \$1,000,000 order	4,850	263,505	913,013
Units displayed at the best quote	225	4,945	1,673,822
Order size / displayed size	21.5	53.3	0.55

Table 1: The three markets, September 2019. Units are shares for the stocks and euros for EUR/USD.

The three assets differ on three separate dimensions (Table 1).

- **Cost of trading.** EUR/USD is the cheapest (0.27 bps quoted spread) and AAPL is close behind (0.73 bps). GPRO is about 30 times more expensive than AAPL (22.5 bps): it trades near \$4.50, so the one-cent minimum tick alone is 22 bps of the price and its spread is pinned at one tick.
- **Tick frequency.** AAPL’s best quote changes price 8.8 times per second, EUR/USD 1.2 times, GPRO once every 25 seconds. The quote a delayed order meets differs from the one the signal saw 62% of the time after 1 second on AAPL, 55% on EUR/USD and 1.1% on GPRO.
- **Size available.** The \$1,000,000 order is 22 times the size displayed at the best quote on AAPL and 53 times on GPRO, but only half of it on EUR/USD.

Limits of the data. The paper and the discussion session use a price-impact price, the average price paid when the order walks the book. MBP-1 has no depth, so every order is assumed to fill in full at the best quote. The last line of Table 1 shows that this is realistic for the currency pair and optimistic for the two stocks. Under this assumption returns in percent do not depend on the order size: they correspond to the paper’s smallest capacity level, and the dollar figures scale them to \$1,000,000. Spot FX has no public trade tape, so the volume input of on-balance volume is tick volume for EUR/USD (one unit for each quote that moves the mid). Dukascopy is one venue, so its tick frequency understates the whole EUR/USD market.

The three signals

Inputs are sampled at each one-minute interval change t . Let m_t be the mid quote and $\bar{x}_t(n)$ the average of the last n values of a series x .

Moving average, MA (paper, eq. 3). With a short window s , a long window l and a band b ,

$$\text{MA}_t = \begin{cases} +1 & \text{if } \bar{m}_t(s) > (1+b) \bar{m}_t(l), \\ -1 & \text{if } \bar{m}_t(s) < (1-b) \bar{m}_t(l), \\ 0 & \text{otherwise.} \end{cases}$$

Baseline: $s = 5$, $l = 30$, $b = 5$ bps, the smallest non-zero band of the paper’s grid, so that the rule also trades on EUR/USD.

On-balance volume, OBV (paper, eq. 6). The running sum since the open of trade volume v_j signed by the tick rule, auction crosses excluded:

$$obv_t = \sum_{j \leq t} \text{sign}(p_j - p_{j-1}) v_j.$$

The rule is long if $\overline{obv}_t(s) > \overline{obv}_t(l) + \text{band}$ and short if $\overline{obv}_t(s) < \overline{obv}_t(l) - \text{band}$. Baseline: $s = 5$, $l = 30$, $b = 0.5$. There are two departures from the paper. Volume is added on an uptick and subtracted on a downtick, the standard convention; the equation as printed in the paper has the opposite sign. And the paper’s band is proportional to the level of OBV; because OBV restarts at zero every day and changes sign, our band is b times the average one-minute volume over the long window.

Persistent quote imbalance, IMB (our own signal). At every instant the imbalance of the best quotes is

$$\frac{q^{bid} - q^{ask}}{q^{bid} + q^{ask}},$$

with q the size at the best bid and ask. I_t is its time-weighted average over the minute ending at t . The rule is long if $I > \theta$ in each of the last k minutes, short if $I < -\theta$ in each of them, flat otherwise. Baseline: $k = 2$, $\theta = 0.3$. The idea is that a queue that stays heavier on the bid for whole minutes signals buying pressure that has not yet moved the price; requiring persistence filters the flickering that dominates instantaneous imbalance.

Parameters. The baseline lookbacks and bands are values of the paper’s 60-second settings (its Appendix A). The same parameters are used for the three assets and were not tuned to improve the P&L. Lookback windows start at the 09:30 open, so the baseline MA and OBV give their first signal at 10:00 and IMB at 09:40. The paper warms its rules up on pre-market quotes; we do not, because GPRO’s pre-market book is too thin to give a meaningful mid.

Rule universe. One rule per signal is enough for the 27 reports, but not to measure an effect as small as the cost of delay. For that part we follow the paper and use many rules: every combination of its 60-second grid for MA and OBV (288 rules each) and 25 IMB rules, 601 rules per asset in total (Appendix A).

The 27 profit-and-loss reports

Asset	Signal	Round trips	Total return (%)			P&L (\$)		
			0 ms	100 ms	1 s	0 ms	100 ms	1 s
AAPL	MA	304	−5.75	−5.88	−6.03	−60,963	−62,273	−63,922
	OBV	103	−0.89	−0.81	−0.90	−9,603	−8,827	−9,700
	IMB	42	−0.75	−0.77	−0.73	−7,954	−8,196	−7,760
GPRO	MA	299	−56.62	−56.62	−57.24	−682,477	−682,477	−690,382
	OBV	150	−26.52	−26.52	−26.10	−310,935	−310,935	−305,665
	IMB	618	−109.10	−108.87	−109.82	−1,293,808	−1,291,173	−1,301,713
EUR/USD	MA	56	−0.19	−0.19	−0.18	−1,899	−1,945	−1,835
	OBV	158	−1.48	−1.45	−1.36	−14,846	−14,617	−13,695
	IMB	53	−0.27	−0.28	−0.28	−2,730	−2,766	−2,803

Table 2: The 27 profit-and-loss reports. Total return is the sum of the simple round-trip returns; P&L is in dollars for the fixed number of units.

Asset	Signal	P&L at mid (\$)	Spread paid (\$)	Net bps / trip	Wins (%)	Longs	Shorts	Holding (min)	Time in position (%)
AAPL	MA	-37,878	23,086	-1.9	33	150	154	13.8	57
	OBV	-1,334	8,269	-0.9	42	17	86	12.2	17
	IMB	-4,971	2,983	-1.8	31	35	7	1.3	1
GPRO	MA	+108,037	790,514	-18.9	18	156	143	19.1	77
	OBV	+84,321	395,257	-17.7	17	69	81	10.7	22
	IMB	+339,921	1,633,729	-17.7	9	360	258	3.8	32
EUR/USD	MA	-347	1,552	-0.3	27	30	26	9.8	7
	OBV	-10,564	4,282	-0.9	23	85	73	9.6	20
	IMB	-1,516	1,214	-0.5	25	25	28	1.8	1

Table 3: Where the P&L comes from, with instantaneous execution. *P&L at mid* values the same trades at the mid quote, so *Spread paid* is the difference with the actual P&L. *Holding* is the average length of a position.

- **Every strategy loses money net of the spread** (Table 2). Total returns range from -0.19% (EUR/USD, MA, 56 round trips) to -109% (GPRO, IMB, 618 round trips). A book below zero is possible because the same number of units is traded on every signal without compounding: 618 round trips at about 18 bps each cost more than the book. The paper studies speed, not profitability, and most of its rules lose as well (58% of those that trade on a given day).
- **On GPRO the loss is the spread** (Table 3). Valued at the mid, the three GPRO strategies make money: +\$108,037 for MA, +\$84,321 for OBV and +\$339,921 for IMB. The spread paid is \$0.4 to \$1.6 million. The signals predict the direction of the next move, but not by the 22.5 bps a round trip costs.
- **On AAPL the signal itself loses.** AAPL MA loses \$37,878 at the mid out of \$60,963 in total: a trend rule on one-minute mids buys after the rise and the price partly reverts. The spread is a secondary cost there.
- **Holding times.** MA positions last 10 to 19 minutes on average, OBV positions 10 to 12 minutes, IMB positions 1 to 4 minutes. MA trades about 300 times on each stock and 56 times on EUR/USD, whose one-minute volatility (1.1 bps) rarely carries the short average 5 bps away from the long one. IMB trades far more on GPRO (618 round trips) than on AAPL (42) or EUR/USD (53): GPRO's queues are large and persistently one-sided.
- **The three delay columns are almost identical.** For each strategy the delays differ by far less than the strategy loses.

Cost of delay of the nine strategies

Table 4 gives the cost of delay at the two required delays, with two measures of uncertainty: the Wilcoxon signed-rank test on the 20 daily differences, which is the paper's test, and a 90% interval obtained by resampling the 20 trading days 5,000 times. Figure 1 extends the delays to the whole range from 0 to 1,000 ms.

Asset	Signal	100 ms		1 second			Wilcoxon p
		Cost (%)	90% interval	Cost (%)	90% interval	Cost (\$)	
AAPL	MA	−0.12	[−0.39, 0.09]	−0.27	[−0.80, 0.26]	−2,958	0.37
	OBV	+0.07	[0.01, 0.14]	−0.01	[−0.20, 0.18]	−97	0.90
	IMB	−0.02	[−0.08, 0.02]	+0.02	[−0.05, 0.09]	+194	0.54
GPRO	MA	0.00	[−0.44, 0.44]	−0.62	[−2.31, 0.68]	−7,905	0.81
	OBV	0.00	[0.00, 0.00]	+0.42	[0.00, 0.87]	+5,270	n/a
	IMB	+0.23	[−0.40, 0.86]	−0.72	[−2.16, 0.57]	−7,905	0.28
EUR/USD	MA	−0.005	[−0.015, 0.007]	+0.006	[−0.026, 0.043]	+64	0.85
	OBV	+0.023	[0.002, 0.053]	+0.115	[−0.015, 0.322]	+1,150	0.57
	IMB	−0.004	[−0.007, −0.001]	−0.007	[−0.026, 0.013]	−73	0.45
Mean of the nine		+0.02	[−0.07, 0.10]	−0.12	[−0.42, 0.14]		

Table 4: Cost of delay of the nine strategies: total return with the delay minus total return at 0 ms, in percentage points. Negative means the delay hurts. The Wilcoxon test is not reported when fewer than six days differ.

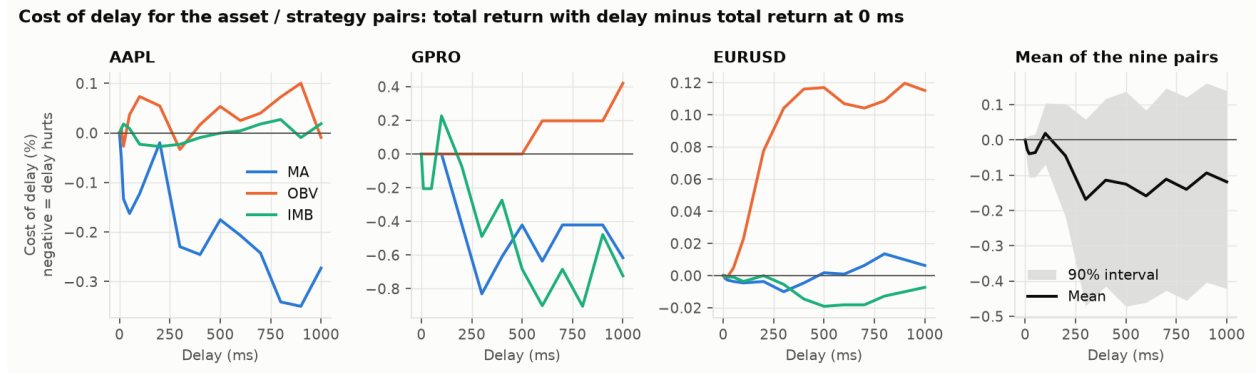


Figure 1: Cost of delay from 0 to 1,000 ms for each asset and signal, and mean of the nine pairs with its 90% interval.

- **At 100 ms the cost is within 0.23 percentage points for all nine pairs**, with no consistent sign: negative for AAPL MA (−0.12%), positive for GPRO IMB (+0.23%) and AAPL OBV (+0.07%), zero for GPRO MA and OBV, whose execution prices almost never change within 100 ms. The mean over the nine pairs is +0.02%.
- **At 1 second the less liquid stock pays the most.** GPRO IMB loses 0.72 percentage points and GPRO MA 0.62 (−\$7,905 each). AAPL MA loses 0.27 (−\$2,958) and EUR/USD MA is unchanged.
- **The OBV strategies gain from delay** at 1 second on GPRO (+0.42%) and EUR/USD (+0.12%).
- **The mean curve is negative at every delay from 200 ms**, between −0.05% and −0.17%, and stands at −0.12% at 1 second. This is the shape of the discussion-session example: a small, noisy effect at short delays, a steady cost at longer ones.

- **None of this is statistically distinguishable from zero.** The 90% interval of the mean is -0.42% to $+0.14\%$ at 1 second, and every 1-second interval in Table 4 contains zero or touches it. The only intervals that exclude zero are three small effects at 100 ms. On GPRO a single one-cent difference on one execution moves the total return by 0.22%.

One strategy over one month places 84 to 1,236 orders. That is not enough to measure an effect of a few hundredths of a basis point per order, which is why the paper works with thousands of rules. The next section does the same.

Importance of speed over a 601-rule universe

All rules (Figure 6 of the paper)

As in the paper, each day the average return across rules with delay d is compared with the average under instantaneous execution, in percent of the latter, and the daily figures are averaged over the 20 days. Rules with a zero return on a day are excluded; on an average day 414 rules trade on AAPL, 547 on GPRO and 303 on EUR/USD.

Asset	Delay (ms)	Mean of daily ratios (%)	Median (%)	Bps per rule-day	Wilcoxon p
AAPL	100	+1.23	+0.10	0.00	0.60
	200	+1.37	+0.13	-0.08	0.93
	500	+3.09	-1.10	-0.32	0.35
	1,000	+0.92	-2.26	-0.77	0.064
GPRO	100	-0.10	0.00	-0.16	0.64
	200	-0.19	0.00	-0.29	0.28
	500	-0.15	0.00	-0.21	0.84
	1,000	-0.49	-0.58	-0.86	0.11
EUR/USD	100	-0.90	+0.08	+0.01	0.81
	200	+3.74	+0.56	+0.08	0.004
	500	+3.53	+1.06	+0.12	0.004
	1,000	+3.15	+2.43	+0.17	0.002

Table 5: Importance of speed for all rules. The first column is the paper’s statistic (mean over days of the relative change in the average return); the second is the median of the same daily figures; the third is the change in the average return in bps per rule-day, which does not depend on a ratio, with its Wilcoxon test.

- **At 100 ms nothing is distinguishable from zero** on any asset.
- **GPRO declines with delay:** -0.10% at 100 ms and -0.49% at 1 second, or 0.86 bps per rule-day ($p = 0.11$).
- **AAPL loses at 1 second, but the paper’s statistic hides it.** The mean of daily ratios is positive ($+0.9\%$) because a few days on which the average return is close to zero dominate it. The median is -2.3% , and in return terms a 1 second delay costs 0.77 bps per rule-day ($p = 0.06$).
- **On EUR/USD delay helps from 200 ms:** $+0.17$ bps per rule-day at 1 second ($p = 0.002$).

These averages mix rules that make money with rules that lose money. The paper shows that the two behave differently, and warns that separating them creates a bias.

Winning and losing rules, and the selection bias (Figure 7 of the paper)

Selecting the rules with a positive return at 0 ms selects trades whose first instants went well. Delaying them removes part of that return mechanically, even if the signal contains no information. The paper measures this selection bias by applying the signals generated on one day to the prices of another, random day: these random rules trade as often and hold as long as the real ones but know nothing about the day they trade. We do the same 5,000 times. An actual value below the 5th percentile of the random rules is a loss that selection cannot explain: a one-sided test at the 5% level, as in the paper.

Importance of speed for winning and losing rules against the selection bias (paper Figure 7)
ring = actual rules below the 5th percentile of the random rules

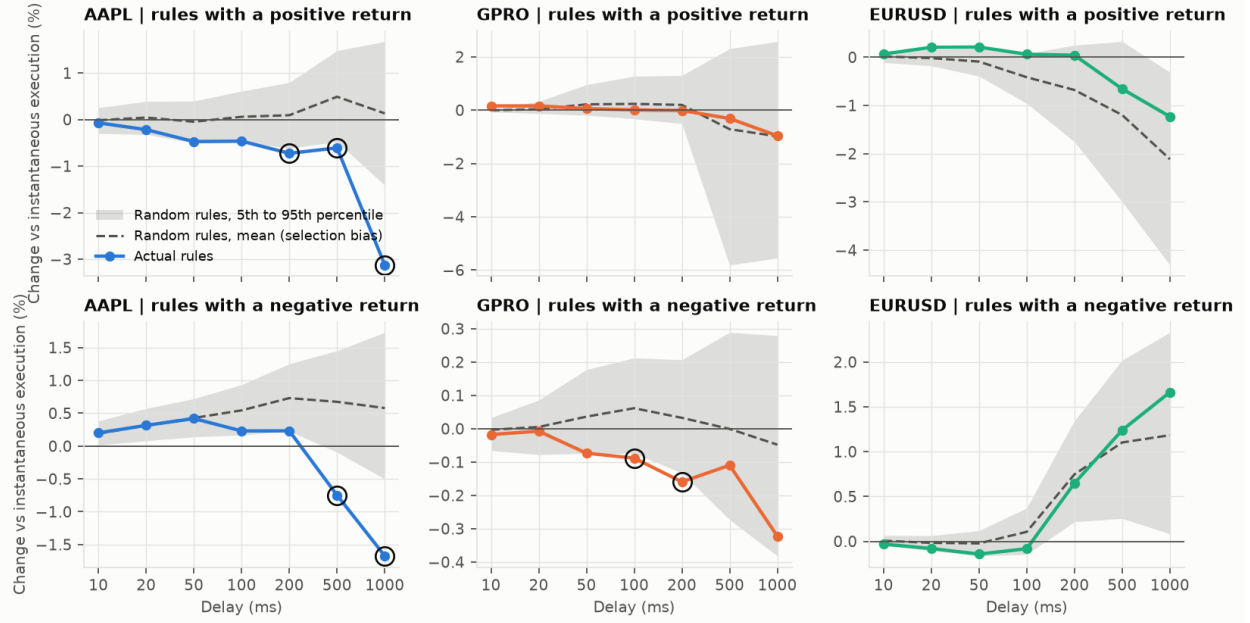


Figure 2: Importance of speed for rules with a positive (top) and negative (bottom) return at 0 ms. Dashed line: mean of the random rules, which is the selection bias. Band: their 5th to 95th percentile. Ring: actual rules below the band.

	Delay (ms)	AAPL			GPRO			EUR/USD		
		Actual (%)	Random rules	p	Actual (%)	Random rules	p	Actual (%)	Random rules	p
Winning rules	100	-0.46	[-0.49, 0.61]	0.058	+0.01	[-0.32, 1.27]	0.42	+0.06	[-0.96, 0.07]	0.94
	200	-0.73*	[-0.61, 0.80]	0.034	-0.01	[-0.51, 1.30]	0.39	+0.04	[-1.75, 0.25]	0.89
	500	-0.61*	[-0.48, 1.48]	0.030	-0.31	[-5.81, 2.30]	0.54	-0.66	[-3.00, 0.32]	0.69
	1,000	-3.13*	[-1.41, 1.68]	< 0.001	-0.96	[-5.55, 2.57]	0.47	-1.23	[-4.31, -0.31]	0.76
Losing rules	100	+0.23	[0.17, 0.93]	0.091	-0.09*	[-0.07, 0.21]	0.034	-0.08	[-0.15, 0.37]	0.11
	200	+0.23	[0.23, 1.25]	0.052	-0.16*	[-0.13, 0.21]	0.033	+0.65	[0.22, 1.34]	0.41
	500	-0.75*	[-0.09, 1.45]	0.001	-0.11	[-0.27, 0.29]	0.27	+1.24	[0.26, 2.02]	0.62
	1,000	-1.67*	[-0.50, 1.73]	< 0.001	-0.32	[-0.38, 0.28]	0.087	+1.66	[0.08, 2.33]	0.76

Table 6: Importance of speed (%) for winning and losing rules. *Random rules* is the 5th to 95th percentile of the same statistic for the random rules; p is the share of the 5,000 draws at or below the actual value. A star marks $p < 0.05$.

- **AAPL, winning rules: speed matters.** Delay lowers the return of profitable rules by 0.5%

at 100 ms, 0.7% at 200 ms and 3.1% at 1 second. Random rules lose nothing (+0.1% at 1 second, band -1.4% to $+1.7\%$), so this is not a selection effect: $p = 0.06$ at 100 ms, 0.03 at 200 ms and at 500 ms, below 0.001 at 1 second. This is the paper’s main result, with the same threshold of about 200 ms.

- **AAPL, losing rules.** Up to 200 ms they behave like random rules. At 500 ms and 1 second they lose more (-1.7% at 1 second against $+0.6\%$ for random rules, $p < 0.001$). The paper finds no effect of speed on losing rules; on AAPL a 1 second delay hurts whatever the rule’s result.
- **GPRO, winning rules: no evidence.** The -1.0% at 1 second equals the selection bias (-1.0% , band -5.5% to $+2.6\%$). Only 67 rules a day make money on GPRO, 12% of those that trade, so the band is wide. Losing rules lose slightly more than random ones at 100 and 200 ms (-0.09% and -0.16% , $p = 0.03$).
- **EUR/USD: the raw numbers are selection.** Winning rules lose 1.2% at 1 second, but random rules lose 2.1% (band -4.3% to -0.3%). Losing rules gain 1.7%, and random ones gain 1.2%. Neither is evidence of an effect of speed.

Without the random rules we would have concluded that delay hurts winning rules on all three assets. Beyond selection, it only does on the liquid stock.

Comparison with the paper

Delay (ms)	Paper, Jan–Sep 2009			This project, Sep 2019		
	SPY	QQQQ	IWM	AAPL	GPRO	EUR/USD
100	−0.07	−0.07	−0.07	−0.46	+0.01	+0.06
200	−0.22*	−0.26	−0.15	−0.73*	−0.01	+0.04
500	−0.39*	−1.01*	−0.62*	−0.61*	−0.31	−0.66
1,000	−1.05*	−1.91*	−1.34*	−3.13*	−0.96	−1.23

Table 7: Importance of speed (%) for rules with a positive return, 60-second signals. Paper: Table B.1 of its internet appendix (186 trading days, 27,424 rules, orders of 100 shares). A star marks a loss that is significant against the selection bias.

The liquid stock reproduces the paper (Table 7): no significant effect at 100 ms, a significant loss from 200 ms, and a loss at 1 second (-3.1%) of the same order as the paper’s ETFs (-1.0% to -1.9%), somewhat larger. Ten years later, and on a single stock instead of an index ETF, the importance of speed has the same shape. The less liquid stock and the currency pair show losses of a similar size at 1 second, which 20 days cannot separate from the selection bias.

Effect of the delay per order

The split between winning and losing rules conditions on performance. The following measure does not. For every order of every rule, the effect of executing it h later is

$$e_h = -\delta \frac{p_{t+h} - p_t}{p_t},$$

with $\delta = +1$ and p the ask for a buy, $\delta = -1$ and p the bid for a sell. It is negative when the price moves in the direction of the order during the delay. Summed over the orders of a strategy it is that strategy’s cost of delay, so this is the same quantity measured order by order. Pooling the 601

rules gives 9,000 to 120,000 orders per signal family and asset. The 90% intervals resample trading days, which preserves the dependence between rules that trade at the same instant.

Asset	Signal	Rules	Orders	100 ms		1 second		Return at 0 ms
				Bps / order	90% interval	Bps / order	90% interval	
AAPL	MA	288	83,616	-0.002	[-0.012, 0.007]	-0.070*	[-0.113, -0.032]	-23.6
	OBV	288	59,390	-0.001	[-0.013, 0.010]	-0.027	[-0.074, 0.012]	-10.4
	IMB	25	11,702	+0.006	[0.000, 0.011]	+0.015	[-0.009, 0.037]	-14.1
GPRO	MA	288	121,602	-0.006	[-0.025, 0.010]	-0.058*	[-0.109, -0.007]	-251.2
	OBV	288	78,714	-0.007	[-0.031, 0.017]	+0.003	[-0.045, 0.056]	-159.3
	IMB	25	25,622	-0.017	[-0.043, 0.005]	-0.107*	[-0.180, -0.040]	-461.5
EUR/USD	MA	288	39,138	-0.0005	[-0.0019, 0.0009]	+0.0029	[-0.0026, 0.0086]	-6.1
	OBV	288	90,128	+0.0010	[-0.0002, 0.0023]	+0.0104*	[0.0061, 0.0151]	-8.4
	IMB	25	9,438	-0.0011*	[-0.0020, -0.0003]	-0.0021	[-0.0058, 0.0016]	-6.1

Table 8: Effect of the delay per order, in bps of the price, over the rule universe. Negative means the delay hurts. A star marks a 90% interval that excludes zero. *Return at 0 ms* is the average return of the family’s rules in bps per rule-day.

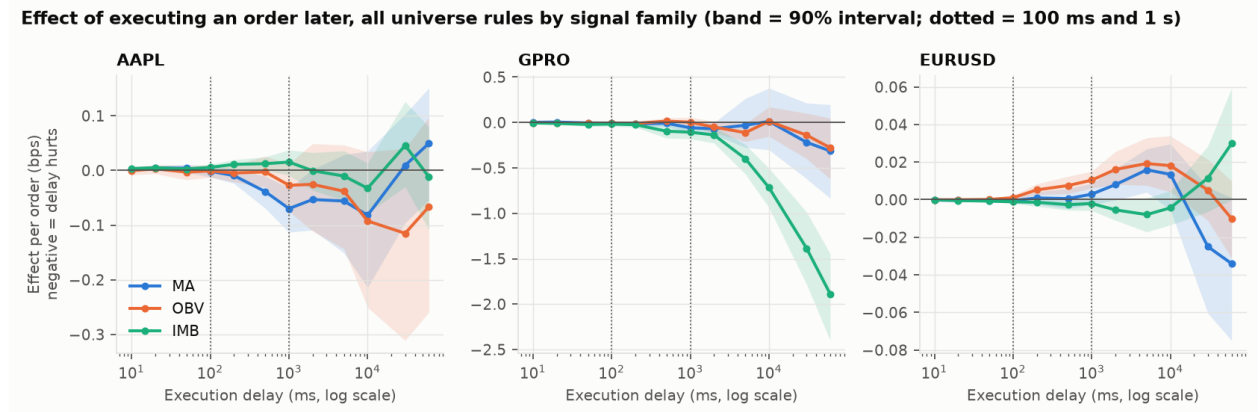


Figure 3: Effect of executing an order later, from 10 ms to 60 seconds, by signal family. Bands are 90% intervals.

- **100 ms costs nothing measurable.** Every family on every asset is within 0.02 bps per order, and only one interval excludes zero (EUR/USD IMB, -0.001 bps).
- **1 second has a cost on the two stocks** (Table 8): 0.07 bps per order for MA rules on AAPL, 0.06 bps for MA rules on GPRO and 0.11 bps for IMB rules on GPRO.
- **The cost is the signal’s short-run predictive power** (Figure 3). After an IMB order on GPRO the price keeps moving in the order’s direction: 0.4 bps after 5 seconds, 1.9 bps after 60 seconds. The signal predicts the next minute, and each second of delay gives up a slice of that move. On AAPL the drift after MA orders builds up over the first second (0.04 bps at 500 ms, 0.07 at 1 second) and is gone after 30 seconds.
- **On EUR/USD delay helps OBV rules**, by 0.01 bps per order at 1 second: the price moves back after their orders. The amount is 4% of the spread.

- **Relative to the spread the cost is small.** 0.07 bps is 19% of AAPL’s half-spread; 0.11 bps is 1% of GPRO’s.

Answers to the three questions of the assignment

(a) Is one asset affected more or less by execution delays, and why?

The two stocks are affected at 1 second; the currency pair is not; nothing is affected at 100 ms.

- **AAPL is the asset where speed matters most clearly.** Its book changes 8.8 times per second, so a delayed order usually meets a different quote, and a one-cent move is worth more than its half-spread. The cost is significant from 200 ms for profitable rules.
- **GPRO pays the most in percent for a single strategy.** Its MA and IMB strategies lose 0.6% and 0.7% at 1 second, against 0.3% for AAPL’s MA, because each change of its quote is a full tick of 22 bps. But its quote changes so rarely that the cost comes from a handful of executions, and it is not statistically measurable for one strategy.
- **EUR/USD is not affected.** Its ticks are frequent but tiny (under 0.1 bp), and its one-minute signals have no short-run predictive power to lose.

A delay costs money when two conditions hold: the quote moves during the delay, and it moves in the direction of the order, which requires a signal that predicts the next move. On AAPL both hold within a second for the MA rules. On GPRO the quote rarely moves, but the imbalance signal predicts its direction when it does. On EUR/USD the quote moves often, by very little, and not in the direction of the orders.

(b) Do the signals perform differently under certain market conditions?

Days are split, asset by asset, into the 10 most and 10 least volatile, by one-minute mid-quote volatility (Table 9).

Asset	Signal	Return at 0 ms (bps / rule-day)		Cost of 1 s (bps / rule-day)		Effect of 1 s per order (bps)	
		High vol	Low vol	High vol	Low vol	High vol	Low vol
AAPL	MA	−27.4	−18.6	−2.14	−0.34	−0.111	−0.017
	OBV	−9.7	−11.2	−1.27	+0.44	−0.080	+0.027
	IMB	−9.8	−18.2	−0.13	+1.43	−0.003	+0.032
GPRO	MA	−256.1	−246.1	−3.16	+0.65	−0.135	+0.032
	OBV	−157.3	−161.4	+0.07	+0.02	+0.004	+0.001
	IMB	−517.7	−406.0	−9.70	−1.40	−0.174	−0.029
EUR/USD	MA	−5.1	−7.3	+0.05	+0.08	+0.0027	+0.0032
	OBV	−9.1	−7.6	+0.33	+0.15	+0.0137	+0.0069
	IMB	−5.8	−6.4	−0.13	−0.01	−0.0039	−0.0002

Table 9: Signals across volatility regimes, over the rule universe. Costs and effects are negative when the delay hurts.

- **The cost of delay is concentrated on volatile days.** MA rules lose 0.11 bps per order to a 1 second delay on AAPL’s volatile days against 0.02 on quiet days, and 0.14 against a small

gain on GPRO. IMB on GPRO loses 0.17 bps per order on volatile days against 0.03. The quote moves further during the delay when the market moves faster.

- **OBV changes sign.** On AAPL a 1 second delay costs OBV rules 0.08 bps per order on volatile days and helps them by 0.03 on quiet days. On GPRO it shows nothing in either regime.
- **Performance itself depends on the regime.** MA rules on AAPL lose more on volatile days (-27 against -19 bps per rule-day): the band is crossed more often and reversed more often. IMB on AAPL loses less on volatile days (-10 against -18). IMB on GPRO loses more (-518 against -406), because it trades more and pays the spread each time.
- **This differs from the paper**, which finds that speed matters more on low-volatility days for SPY.

(c) Does tick frequency affect the importance of delays, and how would we factor it in?

	AAPL	GPRO	EUR/USD
Quote price changes per second	8.80	0.04	1.21
Quote changed after 100 ms / 1 s (%)	25.6 / 62.2	0.2 / 1.1	15.0 / 55.0
Mean absolute mid move after 1 s (bps)	0.48	0.24	0.08
in half-spreads	1.31	0.02	0.62
Absolute effect of 100 ms per round trip (bps)	0.12	0.08	0.01
Absolute effect of 1 s per round trip (bps)	0.38	0.42	0.07
Correlation of daily sensitivity with quote changes	0.48	0.57	0.55
Correlation of daily sensitivity with volatility	0.48	0.69	0.67

Table 10: Tick frequency and sensitivity to delay. The absolute effect is the mean absolute change in return caused by the delay, per round trip, over the rule universe; it measures sensitivity regardless of sign.

Partly: tick frequency sets the probability of a stale quote, and tick size sets what a stale quote costs (Table 10).

- **At 100 ms the fast book is the most exposed:** the absolute effect per round trip is 0.12 bps on AAPL, against 0.08 on GPRO and under 0.02 on EUR/USD.
- **At 1 second GPRO is as sensitive as AAPL** (0.42 against 0.38 bps), although its book is 200 times slower, because each of its ticks is worth 22 bps. EUR/USD is far behind (0.07 bps), although it ticks 30 times faster than GPRO.
- **Within an asset, busier days are more delay-sensitive:** the correlation of daily sensitivity with the number of quote changes per second is 0.48 to 0.57.

How we would factor this into a trading strategy.

1. **Budget latency per asset in units of expected quote move, not in milliseconds.** The relevant number is the probability that the quote changes during the delay times the size of a tick in bps, compared with the signal's expected edge. On AAPL a quarter of the quotes have changed after 100 ms and the measured cost starts around 200 ms; on GPRO 100 ms is free.

2. **Match the signal’s horizon to the achievable latency.** IMB on GPRO pays out over a minute, so a 1 second delay gives up about 6% of its move. The drift after MA orders on AAPL is over within a second: that edge exists only for a sub-second trader.
3. **In large-tick, slow books, do not cross the spread.** GPRO’s signals predict about 2 bps and a round trip costs 22.5 bps. The same signals should be expressed with passive orders: queue position then replaces latency as the scarce resource.
4. **Do not pay for speed on a signal that loses.** Beyond selection, the cost of delay falls on AAPL’s profitable rules. Speed is valuable once the signal predicts the next move; otherwise it just delivers the wrong trade sooner.
5. **Scale speed requirements with volatility.** The cost of a 1 second delay is concentrated on volatile days for both stocks; wider bands or standing aside around scheduled events limit the exposure when the infrastructure cannot be faster.

Limitations

- **One month.** Single-strategy costs of delay cannot be measured, and the pooled results rest on 20 days. With seven delays, three assets and two groups of rules, a few rejections at the 5% level are expected by chance; the AAPL results at 1 second ($p < 0.001$) do not depend on that.
- **Fill at the best quote.** The order is 22 and 53 times the displayed size on the two stocks. True costs, and the cost of arriving after others have taken the best level, are understated there.
- **EUR/USD.** One venue’s quotes, tick volume in place of traded volume, New York hours only.
- **Nasdaq quotes only.** The consolidated best bid and offer may be tighter than the Nasdaq book for AAPL (1.6 cents on average here). GPRO is already at the one-cent minimum.
- **Start of the day.** Lookbacks start at the open, so rules with a 30-minute lookback trade from 10:00 and those with a 60-minute lookback from 10:30.
- **No fees or rebates,** as in the paper.

Appendix A. Rule universe

Family	Parameters	Rules
MA	$s \in \{2, 5, 10, 15, 20, 25, 30\}$, $l \in \{10, 15, 20, 25, 30, 35, 40, 45, 60\}$ $s < l$ (48 pairs), $b \in \{0, 5, 10, 15, 25, 40\}$ bps	288
OBV	the same 48 pairs (s, l) $b \in \{0, 0.1, 0.25, 0.5, 1, 1.5\}$ times the average one-minute volume	288
IMB	$k \in \{1, 2, 3, 4, 5\}$, $\theta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$	25
Total per asset		601

Table 11: Rules of the universe. The lookbacks and the MA bands are the paper’s 60-second settings; its two universal filters are at their neutral values (no confirmation delay, no minimum holding period). Baseline rules: MA (5, 30, 5 bps), OBV (5, 30, 0.5), IMB (2, 0.3).

Appendix B. Additional figures

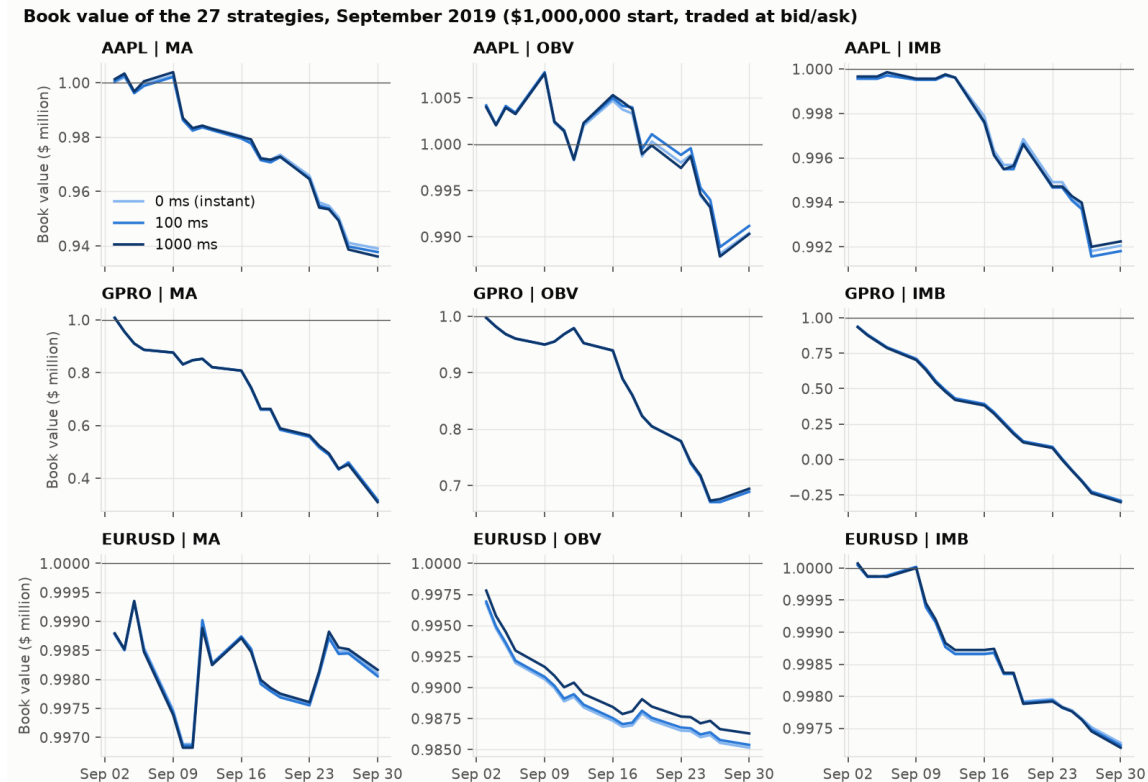


Figure 4: Book value of the 27 strategies over the month.

Importance of speed, all 601 rules per asset (paper Figure 6)

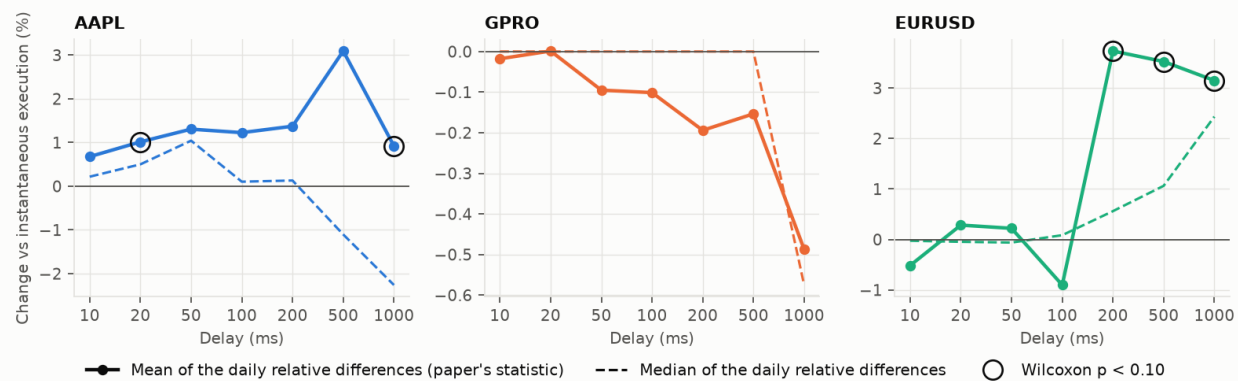


Figure 5: Importance of speed for all rules, at the paper's seven delay levels.

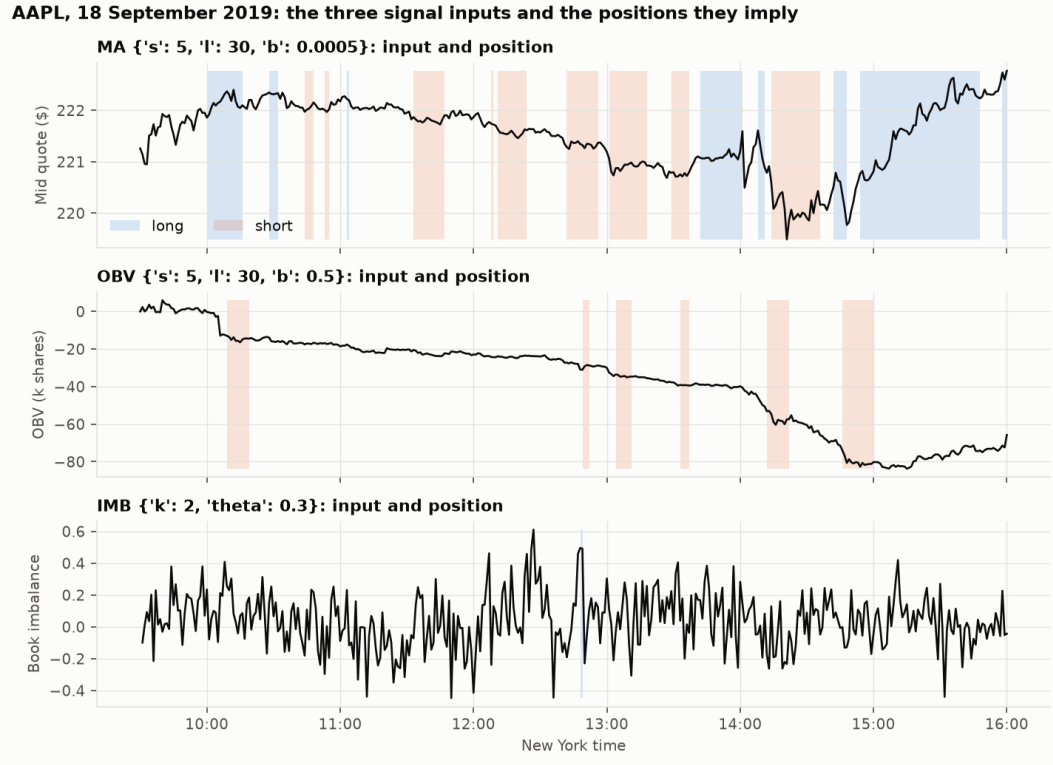


Figure 6: The three signal inputs on AAPL on 18 September 2019 and the positions the baseline rules take.

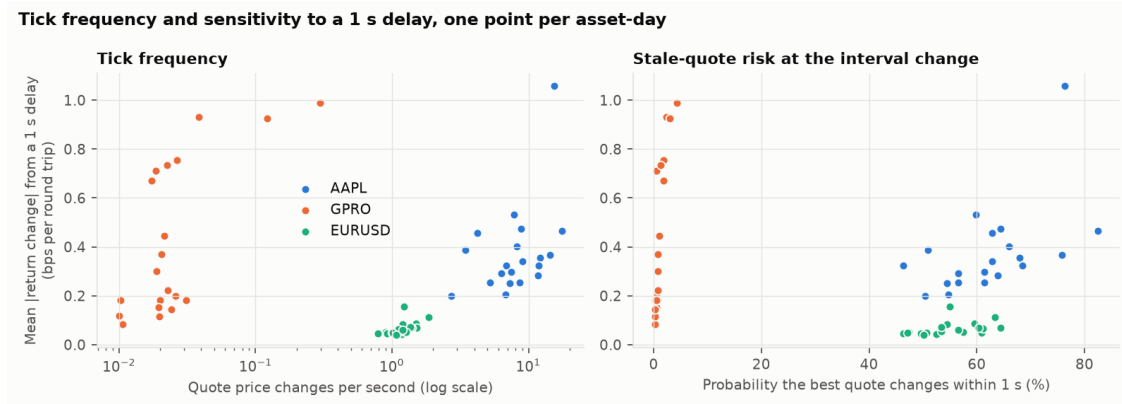


Figure 7: Sensitivity to a 1 second delay against tick frequency, one point per asset-day.

Appendix C. Code and reproduction

Everything is in `Project/Part1_Speed/` of the group repository.

- `MFE230X_project_speed.ipynb`: the executed analysis, with every table and figure of this report.
- `src/speed.py`: data loaders, the three signals, the delayed-execution simulator, the random-rule test and the reporting functions.

- `src/download_fx.py`: downloads the EUR/USD ticks. `src/report_tables.py`: writes the tables of this report from the notebook's outputs.
- `data/`: the Databento and Dukascopy files. `outputs/`: the tables (CSV) and figures.

To reproduce: install `databento`, `jupyter`, `matplotlib`, `numpy`, `pandas` and `scipy`, then execute the notebook from `Part1_Speed/` (about two minutes).

Reference. M. Scholtus and D. van Dijk (2012), *High-Frequency Technical Trading: The Importance of Speed*, Tinbergen Institute Discussion Paper.